A PROOF OF PETER BALA'S FOURTH CONJECTURE ON CUBIC CENTRAL BINOMIAL SUMS

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Abstract. Let $A(N) = \sum_{k=0}^{N} \binom{2k}{k}^3$. We prove Peter Bala's conjecture that n^2 divides $A((n-1)/2)$ whenever n is a product of distinct primes congruent to 3, 5, or 6 modulo 7. More generally, for every such prime p and every positive odd multiple n of p, we prove $p^2 \mid A((n-1)/2)$, without restricting the other prime factors of n. The argument combines Zhi-Hong Sun's prime-level supercongruence with a parameter derivative of a classical Gegenbauer square identity. This derivative annihilates the harmonic correction in every translated half block of the binomial sum. We also give a polynomial proof of the prime-level input and a complete Lean formalization of the squarefree assertion and the stronger odd-multiple theorem.

1. Introduction and main results

The sequence

$$A(N) = \sum_{k=0}^{N} c_k^3, \qquad c_k = \binom{2k}{k}, \qquad N \geq 0, \tag{1}$$

is entry A079727 in the On-Line Encyclopedia of Integer Sequences [1]. In a comment dated July 12, 2024, Peter Bala proposed four congruence statements concerning this sequence. His fourth conjecture is the squarefree divisibility assertion in Corollary 1.2 below. The primes occurring in that conjecture form OEIS A003625; equivalently, they are the primes $p \equiv 3, 5, 6 \pmod 7$.

The relevant prime-level result is already known. Zhi-Hong Sun proved [2, Theorem 3.2] that

$$\sum_{k=0}^{(p-1)/2} c_k^3 \equiv 0 \pmod{p^2} \qquad \text{if } p \equiv 3, 5, 6 \pmod 7. \tag{2}$$

The case $p = 3$ is also immediate from $1+2^3 = 9$. The step required for Bala's conjecture is to propagate this vanishing from the initial half block to all its translates by multiples of p. We establish the following stronger statement.

Theorem 1.1. *Let p be a prime with $p \equiv 3, 5,$ or $6 \pmod 7$. For every positive odd integer n divisible by p,*

$$p^2 \mid A\left(\frac{n-1}{2}\right). \tag{3}$$

Corollary 1.2 (Bala's fourth conjecture)**.** *Let S be a finite set of primes, each congruent to 3, 5, or 6 modulo 7, and put $n = \prod_{p \in S} p$. Then*

$$n^2 \mid A\left(\frac{n-1}{2}\right). \tag{4}$$

The empty set is allowed, with $n = 1$.

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The main ingredient is a harmonic cancellation principle (Lemma 2.1). At any zero modulo p of the truncated cubic binomial polynomial, a particular harmonic-weighted polynomial also vanishes. We obtain this by differentiating a finite Gegenbauer square identity with respect to its parameter, while holding its degree fixed. The weighted polynomial is precisely the correction that arises when c_b^3 is replaced by c_{ap+b}^3 modulo p^2.

Sections 2–4 give the proof. Section 5 describes the accompanying formalization. Appendix A derives the classical square identity, and Appendix B gives a polynomial proof of (2) corresponding to the proof used in Lean. Thus the prime-level result and the classical identity are attributed to the literature, while their required instances are also proved within the formal development.

2. Harmonic cancellation from a square identity

All rational congruences are interpreted in

$$\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}.$$

For an odd prime p, set $h = (p-1)/2$ and write

$$H_j = \sum_{i=1}^{j} \frac{1}{i}, \qquad H_0 = 0.$$

Define

$$F_p(z) = \sum_{k=0}^{h} c_k^3 z^k, \qquad B_p(z) = \sum_{k=0}^{h} c_k^3 z^k (H_{2k} - H_k). \tag{5}$$

Since $2k \le p-1$, all harmonic denominators in (5) are units in $\mathbb{Z}_{(p)}$.

For the rising factorial, write $(a)_k = \prod_{j=0}^{k-1}(a+j)$, with $(a)_0 = 1$. Let $C_m^\lambda(x)$ be the Gegenbauer polynomial defined by

$$(1 - 2xu + u^2)^{-\lambda} = \sum_{m \ge 0} C_m^\lambda(x) u^m,$$

and normalize it by

$$Q_m(\lambda, x) = \frac{m!}{(2\lambda)_m} C_m^\lambda(x). \tag{6}$$

At $\lambda = 1/2$, this is the Legendre polynomial $P_m(x)$. We use the finite square identity

$$Q_m(\lambda, x)^2 = \sum_{k=0}^{m} \frac{(-m)_k (m+2\lambda)_k (\lambda)_k}{(2\lambda)_k (\lambda + \frac{1}{2})_k \, k!} (1-x^2)^k. \tag{7}$$

This is a classical identity; see Sanchez-Ruiz [3, Eq. (16)] and its statement in Brauchart and Grabner [4, proof of Theorem 1]. It is an identity of rational functions in λ with polynomial dependence on x, wherever the displayed denominators are nonzero. A derivation is included in Appendix A.

Lemma 2.1 (Harmonic cancellation)**.** *Let $p \ge 5$ be prime and let $z \in \mathbb{Z}_{(p)}$. If $F_p(z) \equiv 0 \pmod p$, then $B_p(z) \equiv 0 \pmod p$.*

Proof. Use (7) with $m = h$, and denote its kth coefficient by

$$T_k(\lambda) = \frac{(-h)_k (h+2\lambda)_k (\lambda)_k}{(2\lambda)_k (\lambda + \frac{1}{2})_k \, k!}.$$

At $\lambda = 1/2$, the denominators of these coefficients and their parameter derivatives are prime to p. Moreover, $-h \equiv h+1 \equiv 1/2 \pmod p$, and every numerator factor at this parameter is also nonzero

modulo p for $k \le h$. Hence

$$T_k(1/2) \equiv \frac{(\frac{1}{2})_k^3}{(k!)^3} = \frac{c_k^3}{64^k} \pmod{p}. \tag{8}$$

Logarithmic differentiation gives

$$\begin{aligned}\frac{T_k'(1/2)}{T_k(1/2)} &= 2\sum_{j=0}^{k-1}\frac{1}{h+1+j} + \sum_{j=0}^{k-1}\frac{1}{\frac{1}{2}+j} - 3H_k \\ &\equiv 3\sum_{j=0}^{k-1}\frac{1}{\frac{1}{2}+j} - 3H_k \\ &= 6(H_{2k} - H_k) \pmod{p},\end{aligned} \tag{9}$$

where we used $\sum_{j=0}^{k-1}(\frac{1}{2}+j)^{-1} = 2H_{2k} - H_k$. For $k = 0$, the sums are empty and the same formula holds.

Let K be an algebraic closure of $\mathbb{F}_p$, let $\bar{z}$ be the reduction of z, and choose $\xi \in K$ with $\xi^2 = 1 - 64\bar{z}$. Reducing (7) at $\lambda = 1/2$ gives

$$P_h(\xi)^2 = F_p(\bar{z}). \tag{10}$$

The hypothesis implies $P_h(\xi) = 0$, since K is a field. Differentiate (7) with respect to λ, keeping h and x fixed, then reduce modulo p and set $x = \xi$. The left-hand side becomes

$$2P_h(\xi)\left.\frac{\partial Q_h}{\partial \lambda}\right|_{\lambda=1/2,\,x=\xi} = 0.$$

By (8) and (9), the right-hand side is $6B_p(\bar{z})$. Since $p \ge 5$, division by 6 completes the proof. □

3. Translated binomial blocks

The following expansion retains c_{ap} as a prefactor. In particular, no congruence comparing c_{ap} with c_a is needed.

Lemma 3.1 (Binomial translation)**.** *Let p be an odd prime and $a \ge 0$. For $0 \le b \le h$,*

$$c_{ap+b}^3 \equiv c_{ap}^3 c_b^3\big(1 + 6ap(H_{2b} - H_b)\big) \pmod{p^2}. \tag{11}$$

For $h < b < p$, one has $p \mid c_{ap+b}$.

Proof. For $b \le h$, the exact factorial quotient is

$$c_{ap+b} = c_{ap}\frac{\prod_{j=1}^{2b}(2ap+j)}{\prod_{j=1}^{b}(ap+j)^2}.$$

Every factor in the displayed quotient is a unit in $\mathbb{Z}_{(p)}$. Using $(1+pu)^{-1} \equiv 1 - pu \pmod{p^2}$, we obtain

$$\begin{aligned}\frac{\prod_{j=1}^{2b}(2ap+j)}{\prod_{j=1}^{b}(ap+j)^2} &= c_b\frac{\prod_{j=1}^{2b}(1+2ap/j)}{\prod_{j=1}^{b}(1+ap/j)^2} \\ &\equiv c_b\big(1 + 2ap(H_{2b} - H_b)\big) \pmod{p^2}.\end{aligned}$$

Cubing proves (11); the proof does not divide by c_{ap} in the residue ring, so it also covers $p \mid c_{ap}$.

For $h < b < p$, Lucas's binomial congruence gives

$$\binom{2ap+2b}{ap+b} \equiv \binom{2a+1}{a}\binom{2b-p}{b} = 0 \pmod{p},$$

because $0 \le 2b - p < b$. □

Proposition 3.2 (Block propagation)**.** *Let p be an odd prime and $z \in \mathbb{Z}_{(p)}$. If $F_p(z) \equiv 0 \pmod{p^2}$, then for every $a \geq 0$,*

$$\sum_{b=0}^{h} c_{ap+b}^3 z^{ap+b} \equiv 0 \pmod{p^2}, \tag{12}$$

$$\sum_{b=0}^{p-1} c_{ap+b}^3 z^{ap+b} \equiv 0 \pmod{p^2}. \tag{13}$$

Proof. Lemma 3.1 shows that the half-block sum is congruent to

$$z^{ap} c_{ap}^3 \big(F_p(z) + 6apB_p(z)\big) \pmod{p^2}. \tag{14}$$

If $p \geq 5$, Lemma 2.1 annihilates $B_p(z)$ modulo p. If $p = 3$, the factor $6ap = 18a$ is itself divisible by p^2. In both cases (14) vanishes, proving (12). Every remaining term in (13) has $p \mid c_{ap+b}$, so its cube is divisible by p^3. □

4. Proof of the divisibility theorem

Proof of Theorem 1.1. Write $n = pq$ with q positive and odd, and set $a = (q-1)/2$. Then

$$\frac{n-1}{2} = ap + h.$$

Partition the summation interval into a full blocks and one half block:

$$A\left(\frac{n-1}{2}\right) = \sum_{r=0}^{a-1} \sum_{b=0}^{p-1} c_{rp+b}^3 + \sum_{b=0}^{h} c_{ap+b}^3. \tag{15}$$

Sun's theorem (2) gives $F_p(1) \equiv 0 \pmod{p^2}$. Proposition 3.2, applied with $z = 1$, makes every block in (15) divisible by p^2. □

Proof of Corollary 1.2. Every prime in S is odd, so n is odd. For each $p \in S$, Theorem 1.1 gives $p^2 \mid A((n-1)/2)$. The numbers p^2, for $p \in S$, are pairwise coprime; hence their product n^2 divides the same integer. If S is empty, the assertion is $1 \mid A(0) = 1$. □

Remark 4.1. Theorem 1.1 permits repeated factors of p in n, and imposes no condition on the other prime factors of n. Its conclusion is still divisibility by p^2; no stronger valuation is asserted here.

5. Formal verification

The accompanying Lean source (`A79727.lean` in the development workspace) contains a complete proof in Lean 4.33.1, using mathlib and the Formal Conjectures library [5]. It imports the original definitions of the sequence and the eligible primes.

The final declarations are

- `A79727Proof.all_odd_multiples`, proving Theorem 1.1;
- `A79727Proof.conjecture4`, proving Corollary 1.2.

The latter has exactly the type of `OeisA79727.conjecture4`; an executable check at the end of the file compares the complete types, including every hypothesis.

All mathematical inputs needed for these declarations are proved in the same file. In particular, `prime_case` proves the prime-level supercongruence rather than assuming it. The polynomial argument behind this declaration is described in Appendix B. The square identity is implemented for the factorial normalization $G_m(\lambda, x) = m! C_m^{\lambda}(x)$, which has polynomial coefficients and admits a version

with denominators cleared over arbitrary commutative rings. This permits specialization both to characteristic p and to $\mathbb{Z}/p^2\mathbb{Z}$. Formal differentiation in the parameter proves the harmonic cancellation, and `shifted_cube_mod` supplies (11) in the residue ring. The weighted results for arbitrary z in Sections 2 and 3 are proved here on paper; the formal application specializes to $z = 1$.

The file contains no admitted proof or custom axiom. Its final axiom audit checks that only Lean's standard axioms `propext`, `Classical.choice`, and `Quot.sound` occur in the dependencies of the main theorems. The imported repository contains an admitted version of the target; our proof does not depend on that admission. The comparison command reads only the imported declaration's type. Thus importing the problem statement introduces no admitted mathematical premise into the proved result.

Parenthesization of the source statement. The repository uses a compact product expression for its index. To make the grouping explicit for readers, we recommend inserting parentheses around the product, so that the index is visibly

$$\frac{\left(\prod_{p\in S} p\right) - 1}{2}.$$

The declaration `repository_index` proves their equality by `rfl`: the two expressions have the same meaning in Lean. This is a clarification of notation, with no change to the conjecture.

Reproduction. The source statement and definitions were compared with the Formal Conjectures snapshot at commit `df3f12d7bd06feb3f71ae37abae0ca7cb798d9b1`. Within the supplied workspace, the complete check is run by

```
bash scripts/lake.sh env lean
  research/a79727-20261003/A79727.lean
```

where the two displayed lines form a single shell command. In an environment providing the same imports and toolchain, one can instead invoke `lake env lean A79727.lean`. The companion verification record includes the source hash, compiler exit status, and axiom-audit output. Exact arithmetic experiments were also used during development; the proof does not rely on a finite search.

Appendix A. Derivation of the finite square identity

For completeness, we derive (7) as an identity over $\mathbb{Q}(\lambda)$. The generating function for the Gegenbauer polynomials gives

$$(1-x^2)q'' - (2\lambda+1)xq' + m(m+2\lambda)q = 0 \tag{16}$$

for $q(x) = Q_m(\lambda, x)$, with $q(1) = 1$ and $q(-x) = (-1)^m q(x)$. Put $t = 1 - x^2$ and choose the formal branch $x = \sqrt{1-t}$ at $t = 0$. Then $y(t) = q(\sqrt{1-t})$ satisfies

$$t(1-t)y'' + \left(\lambda + \tfrac{1}{2} - (\lambda+1)t\right)y' + \frac{m(m+2\lambda)}{4}y = 0. \tag{17}$$

Because $q(x)^2$ is even, $f(t) = y(t)^2$ is a polynomial of degree at most m.

For an equation $y'' + Py' + Qy = 0$, direct differentiation shows that its square satisfies

$$f''' + 3Pf'' + (P' + 4Q + 2P^2)f' + (2Q' + 4PQ)f = 0.$$

Apply this to (17) and write $\theta = t\, d/dt$. After clearing denominators, the equation becomes

$$\left[\theta(\theta+2\lambda-1)(\theta+\lambda-\tfrac{1}{2}) - t(\theta-m)(\theta+m+2\lambda)(\theta+\lambda)\right]f = 0. \tag{18}$$

Writing $f(t) = \sum_{k\geq 0} a_k t^k$, with $a_0 = 1$, gives

$$a_k = \frac{(k-1-m)(k-1+m+2\lambda)(k-1+\lambda)}{k(k-1+2\lambda)(k-1+\lambda+\frac{1}{2})}a_{k-1}.$$

For generic λ, this recurrence determines

$$a_k = \frac{(-m)_k (m+2\lambda)_k (\lambda)_k}{(2\lambda)_k (\lambda+\frac{1}{2})_k \, k!}.$$

The coefficient at $k = m+1$ vanishes. This proves (7) over $\mathbb{Q}(\lambda)$, and hence at every specialization for which the displayed denominators are nonzero. In particular, specialization and parameter differentiation are valid at $\lambda = 1/2$.

Appendix B. A polynomial proof of the prime-level input

We give a proof of the instance of Sun's theorem used above, following the structure of the formalization. Throughout this appendix, let $p \geq 5$ be a prime with $p \equiv 3, 5, 6 \pmod 7$, let $h = (p-1)/2$, and work initially in $K = \overline{\mathbb{F}_p}$. For $a, b, c \in K$, define

$$\mathcal{H}(a,b,c) = [x^{2h}](x^3 + ax^2 + bx + c)^h.$$

Only polynomial coefficient identities are needed; no elliptic-curve point count is assumed.

B.1. **Translation, scaling, and a quadratic transformation.** If $F \in K[x]$ has degree less than $2p-1$, then

$$[x^{p-1}]F(x+r) = [x^{p-1}]F(x). \tag{19}$$

Indeed, the possible extra terms contain $\binom{j}{p-1}$ for $p \leq j \leq 2p-2$, and all these binomial coefficients are divisible by p. Since $3h < 2p-1$, (19) applies to our cubic powers. It yields

$$\mathcal{H}(a+3r, b+2ar+3r^2, c+br+ar^2+r^3) = \mathcal{H}(a,b,c). \tag{20}$$

Substituting $x \mapsto tx$ also gives, for $t \neq 0$,

$$\mathcal{H}(at, bt^2, ct^3) = t^h \mathcal{H}(a,b,c). \tag{21}$$

The third identity is

$$\mathcal{H}(-2a, a^2-4b, 0) = \mathcal{H}(a,b,0). \tag{22}$$

Here is a coefficient proof. The right-hand side equals $[x^h](x^2+ax+b)^h$. Expand in powers of ax to obtain

$$\sum_{j=0}^{h} \binom{h}{j} a^{h-j} [x^j](x^2+b)^j.$$

For odd j the last coefficient is zero. For $j = 2r$, it equals $\binom{2r}{r} b^r$, which in characteristic p is $\binom{h}{r}(-4b)^r$, since $h = -1/2$ in K. Consequently the displayed sum equals $[x^{2h}](x+a)^h(x^2-4b)^h$. Applying (19) with translation $-a$ transforms the cubic $(x+a)(x^2-4b)$ into $x^3 - 2ax^2 + (a^2-4b)x$, proving (22).

B.2. **The discriminant -7 parameter.** Choose $z \in K$ with $z^2 - z + 2 = 0$, and put $r = z - 4$. Then $r^3 - 35r - 98 = 0$. Apply successively a translation by r, (22), and a translation by $2r$ to the cubic $x^3 - 35x - 98$. The coefficient triples are

$$\begin{aligned}(0, -35, -98) &\longmapsto (3r, 3r^2 - 35, 0)\\ &\longmapsto (-6r, 140 - 3r^2, 0)\\ &\longmapsto (0, -35z^4, -98z^6).\end{aligned}$$

All these triples have the same value of $\mathcal{H}$. Since $z \neq 0$, (21) with $t = z^2$ therefore gives

$$\mathcal{H}(0, -35, -98) = z^{2h} \mathcal{H}(0, -35, -98). \tag{23}$$

Quadratic reciprocity gives

$$\left(\frac{-7}{p}\right) = \left(\frac{p}{7}\right) = -1$$

for the specified residue classes. Thus $x^2 - x + 2$, whose discriminant is -7, has no root in $\mathbb{F}_p$. In particular $z^p \neq z$, so $z^{2h} = z^{p-1} \neq 1$. Equation (23) implies

$$\mathcal{H}(0, -35, -98) = 0. \tag{24}$$

Choose $s \in K$ with $s^2 = -7$. Starting with the triple $(-6s, 1, 0)$, apply (22) and then translate by $-4s$:

$$(-6s, 1, 0) \longmapsto (12s, -256, 0) \longmapsto (0, 80, 128s).$$

The last triple is obtained from $(0, -35, -98)$ by the scaling $t = 4s/7$. Hence (24) gives $\mathcal{H}(-6s, 1, 0) = 0$. The Gegenbauer generating function, reduced modulo p and using $-h = 1/2$ in K, identifies

$$P_h(u) = [x^h](1 - 2ux + x^2)^h = \mathcal{H}(-2u, 1, 0). \tag{25}$$

It follows that $P_h(3s) = 0$, and parity also gives $P_h(-3s) = 0$.

B.3. **Lifting the square to modulus p^2.** Let $R_0 = \mathbb{Z}/p^2\mathbb{Z}$ and $R = R_0[\omega]/(\omega^2 + 63)$. This is a free R_0-module with basis $1, \omega$. The coefficients of P_h have denominators prime to p, so $P_h(\omega)$ is defined in R. Write it as $u + v\omega$ with $u, v \in R_0$. Reducing modulo p and evaluating ω at $3s$ and $-3s$ yields

$$\bar{u} + 3s\bar{v} = 0, \qquad \bar{u} - 3s\bar{v} = 0.$$

Since $p \neq 2, 3, 7$, these imply $\bar{u} = \bar{v} = 0$. Thus $P_h(\omega) \in pR$, and consequently

$$P_h(\omega)^2 = 0 \quad \text{in } R. \tag{26}$$

On the other hand, for $0 \leq k \leq h$,

$$\begin{aligned}(-h)_k(h+1)_k &= \prod_{j=0}^{k-1}\left((j+\tfrac{1}{2})^2 - \frac{p^2}{4}\right)\\ &\equiv (\tfrac{1}{2})_k^2 \pmod{p^2}.\end{aligned}$$

Specialize (7) to $m = h$, $\lambda = 1/2$, and $x = \omega$. Since $1 - \omega^2 = 64$, it gives

$$P_h(\omega)^2 = \sum_{k=0}^{h} \frac{(-h)_k(h+1)_k(\frac{1}{2})_k}{(k!)^3}\, 64^k = \sum_{k=0}^{h} c_k^3 \quad \text{in } R.$$

Together with (26), this proves that the scalar $F_p(1)$ vanishes in R. The scalar map $R_0 \to R$ is injective, so $F_p(1) = 0$ in R_0, establishing (2). The remaining eligible prime $p = 3$ was checked directly in Section 1.

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References

[1] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A079727, including Peter Bala's comment of July 12, 2024, https://oeis.org/A079727; see also entry A003625, https://oeis.org/A003625. Accessed October 3, 2026.

[2] Zhi-Hong Sun, *Congruences concerning Legendre polynomials II*, arXiv:1012.3898v2 [math.NT], 2012, Theorem 3.2.

[3] J. Sanchez-Ruiz, *Linearization and connection formulae involving squares of Gegenbauer polynomials*, Applied Mathematics Letters **14** (2001), no. 3, 261–267.

[4] Johann S. Brauchart and Peter J. Grabner, *Weighted L^2-norms of Gegenbauer polynomials*, manuscript dated March 15, 2021, https://www.math.tugraz.at/~grabner/Publications/L2NormGegenbauer.pdf.

[5] The Formal Conjectures Authors, *Formal Conjectures*, Google DeepMind, https://github.com/google-deepmind/formal-conjectures, file `FormalConjectures/OEIS/79727.lean`, commit `df3f12d7bd06feb3f71ae37abae0ca7cb798d9b1`.

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